

Assignment - II

Q1. Obtain a root, correct to four decimal place of each of the following equation using the bisection method.

Sol: Let $f(x) = x^3 + x^2 - 1 = 0$

we have $f(0) = -1 < 0$, $f(1) = 1 > 0$

$f(2) = 8 + 4 - 1 = 11 > 0$

$f(3) = 27 + 9 - 1 = 35 > 0$

so root lies between 0 and 1.

$$x^3 + x^2 - 1 = 0$$

$$x^2(x+1) - 1 = 0$$

$$x^2 = \frac{1}{x+1}$$

$$\therefore x = \sqrt{\frac{1}{x+1}}$$

Take initial value $= \frac{0+1}{2} = \frac{1}{2} = 0.5 = x_0$

$$x_1 = f(x_0) = \frac{1}{\sqrt{x_0+1}} = \frac{1}{\sqrt{0.5+1}} = 0.81649$$

$$x_2 = f(x_1) = \frac{1}{\sqrt{x_1+1}} = \frac{1}{\sqrt{0.81649+1}} = 0.74196$$

$$x_3 = f(x_2) = \frac{1}{\sqrt{x_2+1}} = \frac{1}{\sqrt{0.74196+1}} = 0.75767$$

$$x_4 = f(x_3) = \frac{1}{\sqrt{x_3+1}} = \frac{1}{\sqrt{0.75767+1}} = 0.75407$$

i) $5x \log_{10} x - 6 = 0$

Here $5x \log_{10} x - 6 = 0$
 $\therefore 5x \log(10x) - 6 = 0$

Let $f(x) = 5x \log_{10} x - 6$

1st iteration

Here $f(1) = -1 < 0$ and $f(2) = 7.0103 > 0$
 $\therefore$ now root lies between 1 and 2

$x_0 = \frac{1+2}{2} = 1.5$

$f(x_0) = f(1.5) = 5 \cdot 1.5 \log(1.5) - 6 = 2.8207 > 0$

2nd iteration

Here $f(1) = -1 < 0$ and $f(1.5) = 2.8207 > 0$
 $\therefore$ now the root lies between 1 and 1.5

$x_1 = \frac{1+1.5}{2} = 1.25$

$f(x_1) = f(1.25) = 5 \cdot 1.25 \log(1.25) - 6$
 $= 0.8557 > 0$

3rd iteration

Here $f(1) = -1 < 0$ and $f(1.25) = 0.8557 > 0$
 $\therefore$ now root lies between 1 and 1.25

$x_2 = \frac{1+1.25}{2} = 1.125$

$$f(x_2) = f(1.125) = 5 \cdot 1.125 \log(11.25) - 6 = -0.0873 < 0$$

4th Iteration

Here $f(1.125) = -0.0873 < 0$ and $f(1.25) = 0.8557 > 0$

$\therefore$ new root lies between 1.125 and 1.25

$$x_3 = \frac{1.125 + 1.25}{2} = 1.1875$$

$$f(x_3) = f(1.1875) = 5 \cdot 1.1875 \log(11.875) - 6 = 0.3806 > 0$$

5th Iteration

Here $f(1.125) = -0.0873 < 0$ and $f(1.1875) = 0.3806 > 0$

$\therefore$ now the root lies between 1.125 and 1.1875

$$x_4 = \frac{1.125 + 1.1875}{2} = 1.1562$$

$$f(x_4) = f(1.1562) = 5 \cdot 1.1562 \log(11.562) - 6 = 0.1458 > 0$$

6th Iteration

Here $f(1.125) = -0.0873 < 0$ and $f(1.1562) = 0.1458 > 0$

$\therefore$ now the root lies between 1.125 and 1.1562

$$x_5 = \frac{1.125 + 1.1562}{2} = 1.1406$$

$$f(x_5) = f(1.1406) = 5 \cdot 1.1406 \log(11.406) - 6 = 0.029 > 0$$

7th Iteration

Here $f(1.125) = -0.0873 < 0$ and $f(1.1406) = 0.029 > 0$

$\therefore$ now root lies between 1.125 and 1.1406

$$x_6 = \frac{1.125 + 1.1406}{2} = 1.1328$$

$$f(x_6) = f(1.1328) = 5 \cdot 1.1328 \cdot \log(1.1328) - 6 = -0.0292 < 0$$

8th Iteration

$$\text{Here } f(1.1328) = -0.0292 < 0 \text{ and } f(1.1406) = 0.0292 > 0$$

$\therefore$ now the root lies between 1.1328 and 1.1406

$$x_7 = \frac{1.1328 + 1.1406}{2} = 1.1367$$

$$f(x_7) = f(1.1367) = 5 \cdot 1.1367 \cdot \log(1.1367) - 6 = 0 < 0$$

Approximate root of the equation $5x \log_{10} x - 6 = 0$ using Bisection method is 1.1367 am.

ii) $x^2 + x - \cos x = 0$

$$\text{Here } x^2 + x - \cos x = 0 \quad \therefore x^2 + x - \cos(x) = 0$$

$$\text{Let } f(x) = x^2 + x - \cos(x)$$

$$\text{Here } f(0) = -1 \quad f(1) = 1.4597$$

First Iteration

$$\text{Here } f(0) = -1 < 0 \text{ and } f(1) = 1.4597 > 0$$

$\therefore$ now root lies between 0 and 1.

$$x_0 = \frac{0+1}{2} = 0.5$$

$$f(x_0) = f(0.5) = (0.5)^2 + 0.5 - \cos(0.5) = -0.1276 < 0$$

2nd Iteration

$$\text{Here } f(0.5) = -0.1276 < 0 \text{ and } f(1) = 1.4597 > 0$$

$\therefore$ now the root lies between 0.5 and 1.

$$x_1 = \frac{0.5 + 1}{2} = 0.75$$

$$f(x_1) = f(0.75) = (0.75)^2 + 0.75 - \cos(0.75) = 0.5808 > 0$$

3rd Iteration

Here $f(0.5) = -0.1276 < 0$ and $f(0.75) = 0.5808 > 0$

$\therefore$ now the root lies between 0.5 and 0.75

$$x_2 = \frac{0.5 + 0.75}{2} = 0.625$$

$$f(x_2) = f(0.625) = (0.625)^2 + 0.625 - \cos(0.625) = 0.2047 > 0$$

4th Iteration

Here $f(0.5) = -0.1276 < 0$ and $f(0.625) = 0.2047 > 0$

$\therefore$ now the root lies between 0.5 and 0.625.

$$x_3 = \frac{0.5 + 0.625}{2} = 0.5625$$

$$f(x_3) = f(0.5625) = (0.5625)^2 + 0.5625 - \cos(0.5625) = 0.033 > 0$$

5th Iteration

Here $f(0.5) = -0.1276 < 0$ and $f(0.5625) = 0.033 > 0$

$\therefore$ now root lies between 0.5 and 0.5625.

$$x_4 = \frac{0.5 + 0.5625}{2} = 0.5312$$

$$f(x_4) = f(0.5312) = (0.5312)^2 + 0.5312 - \cos(0.5312) = -0.0487 < 0$$

6th Iteration

Here $f(0.5312) = -0.0417 < 0$ and $f(0.5625) = 0.033 > 0$

$\therefore$ now the root lies between 0.5312 and 0.5625

$$x_5 = \frac{0.5312 + 0.5625}{2} = 0.5469$$

$$f(x_5) = f(0.5469) = (0.5469)^2 + 0.5469 - \cos(0.5469) = -0.0012 < 0$$

7th Iteration

Here $f(0.5469) = -0.0012 < 0$ and $f(0.5625) = 0.033 > 0$

$\therefore$ now the root lies between 0.5469 and 0.5625

$$x_6 = \frac{0.5469 + 0.5625}{2} = 0.5547$$

$$f(x_6) = f(0.5547) = (0.5547)^2 + 0.5547 - \cos(0.5547) = 0.0123 > 0$$

8th Iteration

Here $f(0.5469) = -0.0012 < 0$ and $f(0.5547) = 0.0123 > 0$

$\therefore$ now the root lies between 0.5469 and 0.5547

$$x_7 = \frac{0.5469 + 0.5547}{2} = 0.5508$$

$$f(x_7) = f(0.5508) = 0.5508^2 + 0.5508 - \cos(0.5508) = 0.002 > 0$$

9th Iteration

Here $f(0.5469) = -0.0012 < 0$ and $f(0.5508) = 0.002 > 0$

$\therefore$ now the root lies between 0.5469 and 0.5508

$$x_8 = \frac{0.5469 + 0.5508}{2} = 0.5488$$

$$f(x_8) = f(0.5488) = (0.5488)^2 + 0.5488 - \cos(0.5488) = -0.003 < 0$$

The root of the equation approx is 0.5488.

2. Use the method of false position to find a real root, correct to four decimal place of the following equation:

i) $x^3 + x^2 + x + 7 = 0$

Let $f(x) = x^3 + x^2 + x + 7$

Here

x	0	-1	-2	-3	-4
$f(x)$	7	6	1	-14	-45

1st Iteration

Here $f(-3) = -14 < 0$ and $f(-2) = 1 > 0$.

$\therefore$ now the root lies between $x_0 = -3$ and $x_1 = -2$.

$$x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$x_1 = -3 - (-14) \cdot \frac{-2 - (-3)}{1 - (-14)} = -2.0667$$

$$f(x_1) = f(-2.0667) = (-2.0667)^3 + (-2.0667)^2 + (-2.0667) + 7 = 0.3775 > 0$$

2nd Iteration

Here $f(-3) = -14 < 0$ and $f(-2.0667) = 0.3775 > 0$

$\therefore$ now the root lies between $x_0 = -3$ and $x_1 = -2.0667$

$$x_3 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} = -3 - (-14) \cdot \frac{-2.0667 - (-3)}{0.3775 - (-14)} = -2.0912$$

$$f(x_3) = f(-2.0912) = (-2.0912)^3 + (-2.0912)^2 + (-2.0912) + 2 = 0.1371 > 0$$

3rd iteration

Here $f(-3) = -14 < 0$ and $f(-2.0912) = 0.1371 > 0$

$\therefore$ now the root lies between $x_0 = -3$ and $x_1 = -2.0912$.

$$x_4 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} = -3 - (-14) \cdot \frac{-2.0912 - (-3)}{0.1371 - (-14)}$$

$$f(x_4) = f(-2.1) = (-2.1)^3 + (-2.1)^2 + (-2.1) + 2 = 0.0491 > 0$$

4th iteration

Here $f(-3) = -14 < 0$ and $f(-2.1) = 0.0491 > 0$

$\therefore$ now root lies between $x_0 = -3$ and $x_1 = -2.1$.

$$x_5 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} = -3 - (-14) \cdot \frac{-2.1 - (-3)}{0.0491 - (-14)}$$

$$f(x_5) = f(-2.1031) = (-2.1031)^3 + (-2.1031)^2 + (-2.1031) + 2 = 0.0175 > 0$$

5th iteration

Here $f(-3) = -14 < 0$ and $f(-2.1031) = 0.0175 > 0$

$\therefore$ now the root lies between $x_0 = -3$ and $x_1 = -2.1031$

$$x_6 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} = -3 - (-14) \cdot \frac{-2.1031 - (-3)}{0.0175 - (-14)}$$

$$x_6 = -2.1043$$

$$f(x_6) = f(-2.1043) = (-2.1043)^3 + (-2.1043)^2 + (-2.1043) + 2 = 0.0062 > 0$$

6th Iteration

Here $f(-3) = -14 < 0$ and $f(-2.1043) = 0.0062 > 0$

$\therefore$ now the root lies between -3 and -2.1043

$$x_7 = -3 - (-14) \cdot \frac{-2.1043 - (-3)}{0.0062 - (-14)} = -2.1047$$

$$f(x_7) = f(-2.1047) = (-2.1047)^3 + (-2.1047)^2 + (-2.1047) + 7 = 0.0022 > 0$$

7th Iteration

Here $f(-3) = -14 < 0$ and $f(-2.1047) = 0.0022 > 0$

$\therefore$ now the root lies between $x_0 = -3$ and $x_1 = -2.1047$

$$x_8 = -3 - (-14) \cdot \frac{-2.1047 - (-3)}{0.0022 - (-14)} = -2.1048$$

$$f(x_8) = f(-2.1048) = (-2.1048)^3 + (-2.1048)^2 + (-2.1048) + 7 = 0.0008 > 0$$

8th Iteration

Here $f(-3) = -14 < 0$ and $f(-2.1048) = 0.0008 > 0$

$\therefore$ now root lies between $x_0 = -3$ and $x_1 = -2.1048$

$$x_9 = -3 - (-14) \cdot \frac{-2.1048 - (-3)}{0.0008 - (-14)} = -2.1048$$

$$f(x_9) = f(-2.1048) = (-2.1048)^3 + (-2.1048)^2 + (-2.1048) + 7 = 0.0003 > 0$$

9th Iteration

Here $f(-3) = -14 < 0$ and $f(-2.1048) = 0.0003 > 0$

$\therefore$ now root lies between $x_0 = -3$ and $x_1 = -2.1048$

$$x_{10} = -3 - (-14) \cdot \frac{-2.1048 - (-3)}{0.0003 - (-14)} = -2.1049$$

$$f(x_{10}) = (-2.1049)^3 + (-2.1049)^2 + (-2.1049) + 7 = 0.0001 > 0.$$

10th iteration

Here $f(-3) = -14 < 0$ and $f(-2.1049) = 0.0001 > 0$.

$\therefore$ new root lies between $x_0 = -3$ and $x_1 = -2.1049$.

$$x_{11} = -3 - (-14) \cdot \frac{-2.1049 - (-3)}{0.0001 - (-14)} = -2.1049.$$

$$f(x_{11}) = f(-2.1049) = (-2.1049)^3 + (-2.1049)^2 + (-2.1049) + 7 = 0 > 0.$$

approx root of the equation is -2.1049 .

ii) $x = 3e^{-x}$.

let $f(x) = x - 3e^{-x}$.

Here	x	0	1	2
	$f(x)$	-3	-0.1036	1.594

1st iteration

Here $f(1) = -0.1036 < 0$ and $f(2) = 1.594 > 0$.

$\therefore$ new root lies between $x_0 = 1$ and $x_1 = 2$.

$$x_1 = x_0 - \frac{f(x_0) \cdot (x_1 - x_0)}{f(x_1) - f(x_0)} = 1 - \frac{(-0.1036) \cdot (2 - 1)}{1.594 - (-0.1036)}$$

$$x_1 = 1.061$$

$$f(x_1) = f(1.061) = 1.061 - 3e^{-1.061} = 0.0228 > 0.$$

2nd Iteration

Here $f(1) = -0.1036 < 0$ and $f(1.061) = 0.0224 > 0$

$\therefore$ now root lies between $x_0 = 1$ and $x_1 = 1.061$.

$$x_3 = 1 - \frac{(-0.1036) \cdot (1.061 - 1)}{0.0224 - (-0.1036)} = 1.0501.$$

$$f(x_3) = f(1.0501) = 1.0501 - 3e^{-1.0501} = 0.000320.$$

3rd Iteration

Here $f(1) = -0.1036 < 0$ and $f(1.0501) = 0.0003 > 0$.

$\therefore$ now the root lies between $x_0 = 1$ and $x_1 = 1.0501$.

$$x_4 = 1 - \frac{(-0.1036) \cdot (1.0501 - 1)}{0.0003 - (-0.1036)} = 1.0499.$$

$$f(x_4) = f(1.0499) = 1.0499 - 3e^{-1.0499} = 0 > 0.$$

approx root of the equation is 1.0499.

(ii) $x \tan x + 1 = 0.$

let $f(x) = x \tan(x) + 1.$

Here $f(0) = 1$ $f(1) = 2.5574$ $f(2) = -3.301.$

1st Iteration

Here $f(1) = 2.5574 > 0$ and $f(2) = -3.3701 < 0.$

$\therefore$ new root lies between $x_0 = 1$ and $x_1 = 2$.

$$x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$= 1 - 2.5574 \cdot \frac{2 - 1}{-3.3701 - 2.5574}$$

$$x_2 = 1.4314$$

$$f(x_2) = f(1.4314) = 1.431 \tan(1.4314) + 1 = 11.2059 > 0$$

2nd Iteration

Here $f(1.4314) = 11.2059 > 0$ and $f(2) = -3.3701 < 0$

$\therefore$ now the root lies between $x_0 = 1.4314$ and $x_1 = 2$.

$$x_3 = x_0 - \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= 1.4314 - \frac{11.2059(2 - 1.4314)}{-3.3701 - 11.2059}$$

$$= 1.8658$$

$$f(x_3) = f(1.8658) = 1.8658 \tan(1.8658) + 1 = -5.089 < 0$$

3rd Iteration

Here $f(1.4314) = 11.2059 > 0$ and $f(1.8658) = -5.089 < 0$

$\therefore$ now root lies between $x_0 = 1.4314$ and $x_1 = 1.8658$.

$$x_4 = x_0 - \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= 1.4314 - \frac{11.2059(1.8658 - 1.4314)}{-5.089 - 11.2059}$$

$$= 1.732$$

$$f(x_4) = f(1.732) = 1.732 \tan(1.732) + 1 = -9.6486 < 0$$

4th Iteration

Here $f(1.4314) = 11.2059 > 0$ and $f(1.732) = -9.6486 < 0$

$\therefore$ now root lies between $x_0 = 1.4314$ and $x_1 = 1.732$.

$$x_5 = x_0 - \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= 1.4314 - \frac{11.2059(1.732 - 1.4314)}{-9.6486 - 11.2059}$$

$$f(x_5) = f(1.593) = 1.593 \tan(1.593) + 1 = -70.836 < 0.$$

5th Iteration

$$\text{Here } f(1.4314) = 11.2059 > 0 \text{ and } f(1.593) = -70.836 < 0$$

$\therefore$ now root lies between $x_0 = 1.4314$ and $x_1 = 1.593$.

$$x_6 = 1.4314 - 11.2059 \cdot \frac{1.593 - 1.4314}{-70.836 - 11.2059}$$

$$x_6 = 1.4535$$

$$f(x_6) = f(1.4535) = 1.4535 \tan(1.4535) + 1 = 13.336 > 0.$$

6th Iteration

$$\text{Here } f(1.4535) = 13.336 > 0 \text{ and } f(1.593) = -70.836 < 0.$$

$\therefore$ now root lies between $x_0 = 1.4535$ and $x_1 = 1.593$.

$$x_7 = 1.4535 - 13.336 \cdot \frac{1.593 - 1.4535}{-70.836 - 13.336}$$

$$x_7 = 1.4756$$

7th Iteration

$$\text{Here } f(1.4756) = 16.4547 > 0 \text{ and } f(1.593) = -70.836 < 0$$

$\therefore$ now the root lies between $x_0 = 1.4756$ and $x_1 = 1.593$

$$x_8 = 1.4756 - 16.4547 \cdot \frac{1.593 - 1.4756}{-70.836 - 16.4547}$$

$$x_8 = 1.4997$$

$$f(x_8) = f(1.4997) = 1.4997 \tan(1.4997) + 1 = 21.464 > 0.$$

3. Find the real root, which lies between 2 and 3 of the equation $x \log_{10} x - 1.2 = 0$ using the methods of bisection and false-position to a tolerance of 0.5%.

ans:-

By False positioning method:

$$\text{Let } f(x) = x \log_{10} x - 1.2$$

$$\text{So that } f(1) = -ve.$$

$$f(2) = -ve \text{ and } f(3) = +ve$$

$\therefore$ the root lies between 2 and 3.

$$\text{Taking } x_0 = 2 \text{ and } x_1 = 3$$

$$f(x_0) = -0.59794$$

$$\text{and } f(x_1) = 0.23236$$

using method of False position

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0) = 2.7210 \dots \textcircled{1}$$

$$f(x_2) = f(2.7210) = -0.0171$$

The root lies between 2.7210 and 3

$$\therefore \text{ Taking } x_0 = 2.7210$$

$$x_1 = 3$$

$$f(x_0) = -0.0171 \text{ and } f(x_1) = 0.2313 \text{ in equation } \textcircled{1} \text{ we get}$$

$$x_3 = 2.720 + \frac{0.271}{0.2313 + 0.0171} \times 0.0171 = 2.74020$$

Representing this process, successive approximations are:

$$x_4 = 2.74024 \text{ and } x_5 = 2.7406$$

$\therefore$ the root is 2.7406.

Using Bisection method:

Taking $f(2) = 2 \log_{10}(2) - 1.2 = -0.59794$ (-ve)
and $f(3) = 3 \log_{10}(3) - 1.2 = 0.23136$ (ve)

Therefore the root lies between 2 and 3.

1st Iteration

$$x_1 = \frac{2+3}{2} = 2.5 \quad \text{and} \quad f(2.5) = 2.5 \log_{10}(2.5) - 1.2 = -0.20517$$

$\therefore$ root lies between 2.5 and 3.

2nd Iteration

$$x_2 = \frac{2.5+3}{2} = 2.75 \quad \text{and} \quad f(2.75) = 2.75 \log_{10}(2.75) - 1.2 = 0.00816$$

$\therefore$ the root lies between 2.5 and 2.75.

3rd Iteration

$$x_3 = \frac{2.5+2.75}{2} = 2.625 \quad \text{and} \quad f(2.625) = 2.625 \log_{10}(2.625) - 1.2 = -0.09978$$

$\therefore$ the root lies between 2.625 and 2.75

4th Iteration

$$x_4 = \frac{2.625+2.75}{2} = 2.6875$$

$$f(2.6875) = 2.6875 \log_{10}(2.6875) - 1.2 = -0.04612$$

$\therefore$ root lies between 2.6875 and 2.75

5th Iteration

$$x_5 = \frac{2.6875+2.75}{2} = 2.71875$$

$$f(2.71875) = 2.71875 \log_{10}(2.71875) - 1.2 = -0.01905 \text{ (-ve)}$$

$\therefore$ root lies between 2.71875 and 2.75

6th approximation

$$x_6 = \frac{2.71875 + 2.75}{2} = 2.734375.$$

$$f(2.734375) = 2.734375 \log_{10}(2.734375) - 1.2 = -0.005456 \text{ (-ve)}$$

$\therefore$ roots lie between 2.71875 and 2.75.

7th approximation

$$x_7 = \frac{2.734375 + 2.75}{2} = 2.74218.$$

$$f(2.74218) = 2.74218 \log_{10}(2.74218) - 1.2 = -0.001337$$

8th approximation

$$x_8 = \frac{2.74218 + 2.75}{2} = 2.74609$$

$$\text{and } f(2.74609) = 2.74609 \log_{10}(2.74609) - 1.2 = -0.004750$$

$\therefore$ Root lies between 2.74609 and 2.75.

4. Use the Newton-Raphson method to obtain a root correct to four decimal place of each of the following equation:

i) $x^2 - 5x + 3 = 0$

Let $f(x) = x^2 - 5x + 3$

$$\frac{d}{dx}(x^2 - 5x + 3) = 2x - 5 \therefore f'(x) = 2x - 5$$

Here, $f(0) = 3 > 0$ and $f(1) = -1 < 0$
 $\therefore$ root lies between 0 and 1.

$$x_0 = \frac{0+1}{2} = 0.5 \quad x_0 = 0.5$$

1st Iteration

$$f(x_0) = f(0.5) = 0.5^2 - 5 \cdot 0.5 + 3 = 0.75$$

$$f'(x_0) = f'(0.5) = 2 \cdot 0.5 - 5 = -4$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{0.75}{-4} = 0.6875$$

2nd Iteration

$$f(x_1) = f(0.6875) = 0.6875^2 - 5 \cdot 0.6875 + 3 = 0.0352$$

$$f'(x_1) = f'(0.6875) = 2 \cdot 0.6875 - 5 = -3.625$$

$$x_2 = 0.6875 - \frac{0.0352}{-3.625} = 0.6972$$

3rd Iteration

$$f(x_2) = f(0.6972) = 0.6972^2 - 5 \cdot 0.6972 + 3 = 0.0001$$

$$f'(x_2) = f'(0.6972) = 2 \cdot 0.6972 - 5 = -3.6056$$

$$x_3 = 0.6972 - \frac{0.0001}{-3.6056} = 0.6972$$

4th Iteration

$$f(x_3) = f(0.6972) = 0.6972^2 - 5 \cdot 0.6972 + 3 = 0$$

$$f'(x_3) = f'(0.6972) = 2 \cdot 0.6972 - 5 = -3.6056$$

$$x_4 = 0.6972 - \frac{0}{-3.6056} = 0.6972$$

The approximate root of the equation is 0.6972.

10) $xe^x = \cos x$.

$$\text{Let } f(x) = xe^x - \cos(x) \quad - \frac{d}{dx} (xe^x - \cos(x)) = e^x + xe^x + \sin(x)$$

$$\frac{d}{dx} (xe^x - \cos(x))$$

$$= \frac{d}{dx} (xe^x) - \frac{d}{dx} (\cos(x)) + \frac{d}{dx} (xe^x) = e^x + xe^x$$

$$= (e^x + xe^x) - -\sin(x)$$

$$= e^x + xe^x + \sin(x)$$

$$\therefore f(x) = e^x + xe^x + \sin(x)$$

Here $f(0) = -1 < 0$ and $f(1) = 2.178 > 0$

$\therefore$ root lies between 0 and 1.

$$x_0 = \frac{0+1}{2} = 0.5 \Rightarrow x_0 = 0.5.$$

1st Iteration

$$f(x_0) = f(0.5) = 0.5e^{0.5} - \cos(0.5) = -0.0532$$

$$f'(x_0) = f'(0.5) = e^{0.5} + 0.5e^{0.5} + \sin(0.5) = 2.9525$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{-0.0532}{2.9525} = 0.518.$$

2nd Iteration

$$f(x_1) = f(0.518) = 0.518e^{0.518} - \cos(0.518) = 0.0008$$

$$f'(x_1) = f'(0.518) = e^{0.518} + 0.518e^{0.518} + \sin(0.518) = 3.0435$$

$$x_2 = 0.518 - \frac{0.0008}{3.0435} = 0.5178.$$

3rd Iteration

$$f(x_1) = f(0.5178) = 0.5178 e^{0.5178} - \cos(0.5178) = 0$$

$$f'(x_2) = f'(0.5178) = e^{0.5178} + 0.5178 e^{0.5178} + \sin(0.5178) = 3.0421$$

$$x_3 = 0.5178 - \frac{0}{3.0421} = 0.5178$$

Approximate root of the equation is 0.5178.

or) $\sqrt[3]{18}$

$$f(x) = 18^{1/3}$$

$$\therefore x^3 = 18 \quad \therefore x^3 - 18 = 0$$

$$\frac{d}{dx} (x^3 - 18) = 3x^2 \quad \therefore f'(x) = 3x^2$$

$$f(0) = -18, f(1) = -17, f(2) = -10, f(3) = 9$$

$$\text{Here } f(2) = -10 < 0 \text{ and } f(3) = 9 > 0$$

$\therefore$ root lies between 2 and 3.

$$x_0 = \frac{2+3}{2} = 2.5 \quad x_0 = 2.5$$

1st Iteration

$$f(x_0) = f(2.5) = 2.5^3 - 18 = -2.375$$

$$f'(x_0) = f'(2.5) = 3 \cdot 2.5^2 = 18.75$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2.5 - \frac{-2.375}{18.75} = 2.6267$$

2nd Iteration

$$f(x_1) = f(2.6267) = 2.6267^3 - 18 = 0.1224$$

$$f'(x_1) = f'(2.6267) = 3 \cdot 2.6267^2 = 20.6981$$

$$x_2 = 2.6267 - \frac{0.1224}{20.6981} = 2.6208$$

3rd Iteration

$$f(x_2) = f(2.6208) = 2.6208^3 - 18 = 0.0003$$

$$f'(x_2) = f'(2.6208) = 3 \cdot 2.6208^2 = 20.6051$$

$$x_3 = 2.6208 - \frac{0.0003}{20.6051} = 2.6207$$

4th Iteration

$$f(x_3) = f(2.6207) = 2.6207^3 - 18 = 0$$

$$f'(x_3) = f'(2.6207) = 3 \cdot 2.6207^2 = 20.6049$$

$$x_4 = 2.6207 - \frac{0}{20.6049} = 2.6207$$

approximate root of the equation is 2.6207

- 5) Compute to four decimal place, the root between 1 and 2 of the equation by (a) method of false position and (b) newton-raphson method

i) $x = e^{-x} \cos x$

a) method of false position

$$\text{Let } f(x) = x - e^{-x} \cos(x)$$

$$\text{Here } f(0) = -1 \quad f(1) = 0.8012$$

1st Iteration

$$\text{Here } f(0) = -1 < 0 \text{ and } f(1) = 0.8012 > 0$$

$\therefore$ root lies between $x_0 = 0$ and $x_1 = 1$

$$x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$= 0 - (-1) \cdot \frac{1 - 0}{0.8012 - (-1)} = 0.5552$$

$$f(x_2) = f(0.5552) = 0.5552 - e^{-0.5552} \cos(0.5552) = 0.0674 > 0$$

2nd iteration

Here $f(0) = -1 < 0$ and $f(0.5552) = 0.0674 > 0$

$\therefore$ new root lies between $x_0 = 0$ and $x_1 = 0.5552$

$$x_3 = 0 - (-1) \cdot \frac{0.5552 - 0}{0.0674 - (-1)} = 0.5201$$

$$f(x_3) = f(0.5201) = 0.5201 - e^{-0.5201} \cos(0.5201) = 0.0043 > 0$$

3rd iteration

Here $f(0) = -1 < 0$ and $f(0.5201) = 0.0043 > 0$

$\therefore$ new root lies between $x_0 = 0$ and $x_1 = 0.5201$

$$x_4 = 0 - (-1) \cdot \frac{0.5201 - 0}{0.0043 - (-1)} = 0.5179$$

$$f(x_4) = f(0.5179) = 0.5179 - e^{-0.5179} \cos(0.5179) = 0.0003 > 0$$

4th iteration

Here $f(0) = -1 < 0$ and $f(0.5179) = 0.0003 > 0$

$\therefore$ new root lies between $x_0 = 0$ and $x_1 = 0.5179$

$$x_5 = 0 - (-1) \cdot \frac{0.5179 - 0}{0.0003 - (-1)} = 0.5178$$

$$f(x_5) = f(0.5178) = 0.5178 - e^{-0.5178} \cos(0.5178) = 0 > 0$$

Approximate root of the equation is 0.5178.

b). newton - raphson method.

$$\text{let } f(x) = x - e^{-x} \cos(x)$$

$$\frac{d}{dx} (x - e^{-x} \cos(x)) = 1 + e^{-x} \cos(x) + e^{-x} \sin(x)$$

$$\therefore f'(x) = 1 + e^{-x} \cos(x) + e^{-x} \sin(x)$$

Here

$$f(0) = -1 \quad f(1) = 0.8012$$

$$\text{Here } f(0) = -1 < 0 \text{ and } f(1) = 0.8012 > 0$$

$\therefore$ root lies between 0 and 1

$$x_0 = \frac{0+1}{2} = 0.5 \quad x_0 = 0.5$$

1st Iteration

$$f(x_0) = f(0.5) = 0.5 - e^{-0.5} \cos(0.5) = -0.0323$$

$$f'(x_0) = f'(0.5) = 1 + e^{-0.5} \cos(0.5) + e^{-0.5} \sin(0.5) = 1.8231$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{-0.0323}{1.8231} = 0.5177$$

2nd Iteration

$$f(x_1) = f(0.5177) = 0.5177 - e^{-0.5177} \cos(0.5177) = 0$$

$$f'(x_1) = f'(0.5177) = 1 + e^{-0.5177} \cos(0.5177) + e^{-0.5177} \sin(0.5177) = 1.8127$$

$$x_2 = 0.5177 - \frac{0}{1.8127} = 0.5178$$

Approximate root of the equation is 0.5178.

ii) $x^3 - 6x^2 + 11x - 6 = 0.$

a) method of false position.

Here $x^3 - 6x^2 + 11x - 6 = 0$

$\therefore x^3 - 6x^2 + 11x - 6 = 0$

Let $f(x) = x^3 - 6x^2 + 11x - 6$

Here $f(1) = 0$

$\therefore$ root of the equation $x^3 - 6x^2 + 11x - 6$ is 1.

b) Newton Raphson method.

$f(x) = x^3 - 6x^2 + 11x - 6$

$\frac{d}{dx} (x^3 - 6x^2 + 11x - 6) = 3x^2 - 12x + 11$

$\Rightarrow \frac{d}{dx} (x^3) - \frac{d}{dx} (6x^2) + \frac{d}{dx} (11x) - \frac{d}{dx} (6)$

$\Rightarrow 3x^2 - 12x + 11 = 0$

$\Rightarrow 3x^2 - 12x + 11$

Here $f(0) = -6$ $f(1) = 0$

$\therefore$ root of the equation is 1.

iii) $3x = 1 + \cos x.$

a) method of false position.

Let $f(x) = 3x - \cos(x) - 1$

$f(0) = -2$ $f(1) = 1.4597.$

1st iteration.

Here $f(0) = -2 < 0$ and $f(1) = 1.4597 > 0$

$\therefore$ now root lies between $x_0 = 0$ and $x_1 = 1$

$$x_2 = x_0 - f(x_0) \cdot \frac{(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= 0 - (-2) \cdot \frac{1 - 0}{1.4597 - (-2)} = 0.5781$$

$$f(x_2) = f(0.5781) = 3 \cdot 0.5781 - \cos(0.5781) - 1 = -0.1033 < 0$$

2nd iteration

Here $f(0.5781) = -0.1033 < 0$ and $f(1) = 1.4597 > 0$

$\therefore$ now root lies between $x_0 = 0.5781$ and $x_1 = 1$.

$$x_3 = 0.5781 - (-0.1033) \cdot \frac{1 - 0.5781}{1.4597 - (-0.1033)}$$

$$x_3 = 0.606$$

$$f(x_3) = f(0.606) = 3 \cdot 0.606 - \cos(0.606) - 1 = -0.0041 < 0$$

3rd iteration

Here $f(0.606) = -0.0041 < 0$ and $f(1) = 1.4597 > 0$

$\therefore$ now root lies between $x_0 = 0.606$ and $x_1 = 1$

$$x_4 = 0.606 - (-0.0041) \cdot \frac{1 - 0.606}{1.4597 - (-0.0041)} = 0.6071$$

$$f(x_4) = f(0.6071) = 3 \cdot 0.6071 - \cos(0.6071) - 1 = -0.0002 < 0$$

Approximate root of the equation is 0.6071.

6. Determine the real root of the equation $x = e^{-x}$, using secant method.

Sol: let $f(x) = x - e^{-x}$
 $f(0) = -1$ $f(1) = 0.6321$.

1st Iteration

$$x_0 = 0 \text{ and } x_1 = 1$$

$$f(x_0) = f(0) = -1 \text{ and } f(x_1) = f(1) = 0.6321.$$

$$\therefore x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$= 0 - (-1) \cdot \frac{1 - 0}{0.6321 - (-1)} = 0.6127.$$

$$\therefore f(x_2) = f(0.6127) = 0.6127 - e^{-0.6127} = 0.0708$$

2nd Iteration

$$x_1 = 1 \text{ and } x_2 = 0.6127$$

$$f(x_1) = f(1) = 0.6321 \text{ and } f(x_2) = f(0.6127) = 0.0708.$$

$$\therefore x_3 = x_1 - f(x_1) \cdot \frac{x_2 - x_1}{f(x_2) - f(x_1)}$$

$$x_3 = 1 - 0.6321 \cdot \frac{0.6127 - 1}{0.0708 - 0.6321} = 0.5638.$$

$$\therefore f(x) = f(0.5638) = 0.5638 - e^{-0.5638} = -0.0052.$$

3rd Iteration

$$x_2 = 0.6127 \text{ and } x_3 = 0.5638$$

$$f(x_2) = f(0.6127) = 0.0708 \quad f(x_3) = f(0.5638) = -0.0052$$

$$x_4 = 0.6127 - 0.0708 \cdot \frac{0.5638 - 0.6127}{-0.0052 - 0.0708}$$

$$= 0.5672.$$

$$\therefore f(x_4) = f(0.5672) = 0.5672 - e^{-0.5672} = 0.$$

Approximate root of the equation is 0.5672.

7. Point out the difference between the regula-falsi and the secant methods for finding a real root of $f(x)=0$. Apply both of the methods to find a real root of the equation $x^3 - 2x - 5 = 0$ to an accuracy of 4 decimal places.

Sol: The regula-falsi method and secant method both are numerical techniques used to find real roots of a given function $f(x)=0$. Here are the key difference:

Regula Falsi method	secant method.
→ It uses a bracketed interval where the function changes sign.	It uses two initial form points to form a secant line.
→ The root is estimated by the intersection of x-axis with the straight line connecting the function values at the interval's endpoints.	The root is estimated by the intersection of the secant line with x-axis.
• converges linearly but can be slower in certain cases.	• converges faster but might fail if the initial points are chosen poorly.

now let's apply both methods to find the real root of the equation.
 $x^3 - 2x - 5 = 0$.

a) regula-falsi method.

$$f(x) = x^3 - 2x - 5$$

$$\text{here } f(0) = -5 \quad f(1) = -6 \quad f(2) = -1 \quad f(3) = 16$$

1st iteration

$$\text{Here } f(0) = -5 < 0 \text{ and } f(3) = 16 > 0$$

$\therefore$ now root lies between $x_0 = 0$ and $x_1 = 3$

$$x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)} = 0 - (-5) \cdot \frac{3 - 0}{16 - (-5)} = 2.0588$$

$$f(x_2) = f(2.0588) = 2.0588^3 - 2 \cdot 2.0588 - 5 = -0.3908 < 0$$

2nd iteration

$$\text{Here } f(2.0588) = -0.3908 < 0 \text{ and } f(3) = 16 > 0$$

$\therefore$ now root lies between $x_0 = 2.0588$ and $x_1 = 3$

$$x_3 = 2.0588 - (-0.3908) \cdot \frac{3 - 2.0588}{16 - (-0.3908)} = 2.0813$$

$$f(x_3) = f(2.0813) = 2.0813^3 - 2 \cdot 2.0813 - 5 = -0.1472 < 0$$

3rd iteration

$$\text{Here } f(2.0813) = -0.1472 < 0 \text{ and } f(3) = 16 > 0$$

$\therefore$ now root lies between $x_0 = 2.0813$ and $x_1 = 3$

$$x_4 = 2.0813 - (-0.1472) \cdot \frac{3 - 2.0813}{16 - (-0.1472)} = 2.0896$$

$$f(x_4) = f(2.0896) = 2.0896^3 - 2 \cdot 2.0896 - 5 = -0.0547 < 0$$

4th Iteration

Here $f(2.0896) = -0.0547 < 0$ and $f(3) = 16 > 0$

$\therefore$ root lies between $x_0 = 2.0896$ and $x_1 = 3$.

$$x_5 = 2.0896 - (-0.0547) \cdot \frac{3 - 2.0896}{16 - (-0.0547)} = 2.0927$$

$$f(x_5) = f(2.0927) = 2.0927^3 - 2 \cdot 2.0927 - 5 = -0.0202 < 0$$

b) Secant method.

$$\text{Let } f(x) = x^3 - 2x - 5.$$

$$f(0) = -5 \quad f(1) = -6 \quad f(2) = -1 \quad f(3) = 16.$$

1st Iteration

$$x_0 = 2 \text{ and } x_1 = 3$$

$$f(x_0) = f(2) = -1 \text{ and } f(x_1) = f(3) = 16.$$

$$\therefore x_2 = x_0 - f(x_0) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$x_2 = 2 - (-1) \cdot \frac{3 - 2}{16 - (-1)} = 2.0588$$

$$\therefore f(x_2) = f(2.0588) = 2.0588^3 - 2 \cdot 2.0588 - 5 = -0.3908.$$

2nd Iteration

$$x_1 = 3 \text{ and } x_2 = 2.0588$$

$$f(x_1) = f(3) = 16 \text{ and } f(x_2) = f(2.0588) = -0.3908.$$

$$\therefore x_3 = \frac{3-16 \cdot 2.0588 - 3}{-0.3908-16} = 2.0813.$$

$$f(x_3) = f(2.0813) = 2.0813^3 - 2 \cdot 2.0813 - 5 = -0.1472.$$

3rd iteration

$$x_2 = 2.0588 \text{ and } x_3 = 2.0813.$$

$$f(x_2) = f(2.0588) = -0.3908$$

$$f(x_3) = f(2.0813) = -0.1472.$$

$$\therefore x_4 = \frac{2.0588 - (-0.3908) \cdot \frac{2.0813 - 2.0588}{-0.1472 - (-0.3908)}}{1}$$

$$x_4 = 2.0948$$

$$\therefore f(x_4) = f(2.0948) = 2.0948^3 - 2 \cdot 2.0948 - 5 = 0.003.$$

4th iteration

$$x_3 = 2.0813 \text{ and } x_4 = 2.0948.$$

$$f(x_3) = f(2.0813) = -0.1472 \text{ and } f(x_4) = f(2.0948) = 0.003.$$

$$\therefore x_5 = \frac{2.0813 - (-0.1472) \cdot \frac{2.0948 - 2.0813}{0.003 - (-0.1472)}}{1}$$

$$x_5 = 2.0945.$$

$$\therefore f(x_5) = f(2.0945) = 2.0945^3 - 2 \cdot 2.0945 - 5 = 0.$$

approx root of the equation is 2.0945.

8. find the cubic polynomial which takes the following values.
 $y(1) = 21$, $y(3) = 120$, $y(5) = 336$, and $y(7) = 720$
 obtain the value of $y(6)$.

Sol: we can use Lagrange interpolation formula for polynomials.

The formula is as follows:

$$p(x) = y_1 \frac{(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_2)(x_1-x_3)(x_1-x_4)} + y_2 \frac{(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_1)(x_2-x_3)(x_2-x_4)} + \dots + y_n \frac{(x-x_1)(x-x_2) \dots (x-x_{n-1})}{(x_n-x_1)(x_n-x_2) \dots (x_n-x_{n-1})}$$

Given values $y(1)=24$, $y(3)=120$, $y(5)=336$ and $y(7)=720$.
we can use the polynomials to construct the polynomial.

$$p(x) = 24 \frac{(x-3)(x-5)(x-7)}{(-3)(-5)(-7)} + 120 \frac{(x-1)(x-5)(x-7)}{(3-1)(3-5)(3-7)} + 336 \frac{(x-1)(x-3)(x-7)}{(5-1)(5-3)(5-7)} + 720 \frac{(x-1)(x-3)(x-5)}{(7-1)(7-3)(7-5)}$$

$$p(x) = 2x^3 - 3x^2 + 4x + 20$$

To find $y(8)$, substitute $x=8$ into the polynomial

$$\begin{aligned} y(8) &= 2(8)^3 - 3(8)^2 + 4(8) + 20 \\ &= 1024 - 192 + 32 + 20 \\ &= 884 \text{ ans} \end{aligned}$$

9. find the missing term in the following table:

x	0	1	2	3	4
y	5	13	23	-	67

Sol: To find the missing term in the table, we can use finite difference method. Calculate the first difference (Δy) and 2nd difference ($\Delta^2 y$).

x	y	Δy	$\Delta^2 y$	
0	5	.	.	$\Delta y_1 = 13 - 5 = 8$, $\Delta y_2 = 23 - 13 = 10$
1	13	8	.	The second diff ($\Delta^2 y$) represent the difference between consecutive first difference:
2	23	10	2	$\Delta^2 y_1 = 10 - 8 = 2$
3	-	-	.	$\Delta^2 y_2 = \Delta^2 y_1 = 2$
4	67	.	.	$- = 23 + 2 = 25$
5	.	.	.	

$\therefore$ The missing term in the table is $y = 25$ for $x = 3$.

10. The table below gives the value of $\tan x$ $[0.10, 0.30]$:

x	0.10	0.15	0.20	0.25	0.30
$y = \tan x$	0.1003	0.1511	0.2027	0.2553	0.3093

find a) $\tan 0.12$ b) $\tan 0.20$ c) $\tan 0.40$ and d) $\tan 0.50$.

sol: The table of difference is:

x	y	Δ	Δ^2	Δ^3	Δ^4
0.10	0.1003	0.0508			
0.15	0.1511	0.0516	0.0008		
0.20	0.2027	0.0526	0.0010	0.0002	
0.25	0.2553	0.0540	0.0017	0.0004	
0.30	0.3093				

a) To find $\tan(0.12)$ we have $0.12 = 0.10 + p(0.05)$, which gives
 $p = 0.4$.

$$\tan(0.12) = 0.1003 + 0.4(0.0508) + \frac{0.4(0.4-1)}{2}(0.0006)$$

$$+ \frac{0.4(0.4-1)(0.4-2)}{6}(0.0000)$$

$$+ \frac{0.4(0.4-1)(0.4-2)(0.4-3)}{24}(0.0002)$$

$$= 0.1205$$

b) To find $\tan(0.26)$ we have $0.26 = 0.30 + p(0.05)$
 $p = -0.8$

$$\tan(0.26) = 0.3093 - 0.8(0.0540) + \frac{-0.8(-0.8+1)}{2}(0.0004)$$

$$+ \frac{-0.8(-0.8+1)(-0.8+2)}{6}(0.0000)$$

$$+ \frac{-0.8(-0.8+1)(-0.8+2)(-0.8+3)}{24}(0.0002)$$

$$= 0.2662$$

c) $\tan(0.40) = \tan 0.30 + (\tan 0.25 - \tan 0.30) \times \frac{0.40 - 0.30}{0.25 - 0.30}$

$$= 0.2353642$$

d) $\tan 0.50 = \tan 0.30 + (\tan 0.40 - \tan 0.30) \times \frac{0.50 - 0.30}{0.40 - 0.30}$

$$= -0.729965889$$